A

Real - Time Research Project

On

Automatic Irrigation Based on Soil Moisture Content Control

(Submitted in partial fulfillment of the requirements for the award of Degree)

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

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April, 2026.

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CERTIFICATE

This is to certify that the project entitled "**AUTOMATIC IRRIGATION BASED ON SOIL MOISTURE CONTENT CONTROL**" being submitted by **Ch. Pavithra (247R1A05U0), B. Bhageerath (247R1A05T2) & M. Ramakanth (257R5A0526)** in partial fulfillment of the requirements for the award of the degree of B.Tech in Computer Science and Engineering to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

Automatic Irrigation Based on Soil Moisture Content Control is a smart agricultural system designed to optimize water usage and improve crop productivity. Traditional irrigation methods rely on manual operation or fixed time schedules, which often lead to over-irrigation or under-irrigation. This results in water wastage, increased labor effort, and reduced efficiency in farming. To overcome these issues, the proposed system uses soil moisture sensors to continuously monitor the moisture level present in the soil and automatically control the irrigation process.

The system consists of key components such as a soil moisture sensor, microcontroller (Arduino/ATmega328), water pump, and control circuitry. The soil moisture sensor detects the water content in the soil and sends the data to the microcontroller. When the moisture level drops below a predefined threshold, the microcontroller activates the water pump to supply water to the field. Once the soil reaches the required moisture level, the system automatically turns off the pump. This ensures that crops receive the right amount of water at the right time without human intervention.

Additionally, the system can be integrated with IoT technology for remote monitoring and control through mobile applications or web platforms. Farmers can monitor soil conditions in real-time and manage irrigation efficiently even from distant locations. This not only saves time but also enhances decision-making in agriculture.

The proposed system helps in conserving water, reducing energy consumption, and minimizing human effort. It also improves crop yield by maintaining optimal soil conditions. This solution is cost-effective, easy to implement, and suitable for various agricultural environments, especially in water-scarce regions.

Overall, the Automatic Irrigation System based on Soil Moisture Control promotes sustainable farming practices by ensuring efficient resource utilization and supporting modern precision agriculture.

LIST OF FIGURES

FIGURE NO	FIGURE NAME	PAGE NO
Figure 3.1	Project Architecture	12
Figure 3.2	Data flow Diagram	15
Figure 4.1	Embedded Development Life Cycle	21
Figure 5.1	Hardware Setup of Automatic Irrigation System	27

LIST OF TABLES

TABLE NO	TABLE NAME	PAGE NO
Table 6.2.1	Soil Moisture Detection	30
Table 6.2.2	Pump Activation (Dry Soil)	30
Table 6.2.3	Pump Deactivation (Wet Soil)	30
Table 6.2.4	Continuous Monitoring	30
Table 6.2.5	System Response	30

TABLE OF CONTENTS

ABSTRACT i
LIST OF FIGURES ii
LIST OF TABLES iii
1. INTRODUCTION 1
1.1 PROJECT PURPOSE 1
1.2 PROJECT FEATURES 2
2. LITERATURE SURVEY 3
2.1 REVIEW OF RELATED WORK 5
2.2 DEFINITION OF PROBLEM STATEMENT 7
2.3 EXISTING SYSTEM 7
2.4 PROPOSED SYSTEM 8
2.5 OBJECTIVES 9
2.6 HARDWARE & SOFTWARE REQUIREMENTS 10
2.6.1 HARDWARE REQUIREMENTS 10
2.6.2 SOFTWARE REQUIREMENTS 10
3. SYSTEM ARCHITECTURE & DESIGN 11
3.1 PROJECT ARCHITECTURE 12
3.2 DESCRIPTION 13
3.3 DATA FLOW DIAGRAM 14
4. IMPLEMENTATION 16
4.1 ALGORITHMS USED 17
4.2 SAMPLE CODE 22
5. **RESULTS & DISCUSSION** 25
6. VALIDATION 28
6.1 INTRODUCTION 28
6.2 TESTCASES 30
6.2.1 UPLOADING DATASET 30
6.2.2 CLASSIFICATION 30
7. CONCLUSION & FUTURE ASPECTS 31
7.1 PROJECT CONCLUSION 32
7.2 FUTURE ASPECTS 33
8. BIBLIOGRAPHY 34
8.1 REFERENCES 35
8.2 GITHUB LINK 36

1. INTRODUCTION

1. INTRODUCTION

Agriculture plays a vital role in the economy and food production, but efficient water management remains a major challenge. Traditional irrigation methods depend on manual operation or fixed time schedules, which often lead to overwatering or insufficient watering of crops. This not only wastes water but also affects crop growth and productivity. With increasing water scarcity and the need for sustainable farming, there is a demand for smart and automated irrigation systems.

The Automatic Irrigation System Based on Soil Moisture Content Control is designed to address these issues by using modern technology. This system uses soil moisture sensors to continuously monitor the water content in the soil and automatically control the irrigation process. When the soil becomes dry, the system turns on the water pump, and when sufficient moisture is reached, it turns it off. This ensures that crops receive the required amount of water at the right time. By reducing human intervention and optimizing water usage, this system improves efficiency, saves resources, and supports better crop yield. It is a cost-effective and practical solution for modern agriculture, especially in regions facing water shortages.

1.1 PROJECT PURPOSE

The main purpose of this project is to develop an efficient and automated irrigation system that can monitor soil moisture levels and supply water to crops only when necessary. This helps in reducing water wastage and ensures optimal utilization of available water resources.

The project aims to eliminate the need for manual intervention in irrigation by using sensors and a microcontroller to control the watering process automatically. By maintaining the right moisture level in the soil, the system supports healthy plant growth and improves agricultural productivity.

Another important purpose of this project is to promote sustainable farming practices by conserving water and reducing energy consumption. It also provides a cost-effective and easy-to-use solution for farmers, making it suitable for both small-scale and large-scale agricultural applications.

Overall, the project is intended to enhance efficiency, reduce labor effort, and contribute to modern smart agriculture systems.

1.2 PROJECT FEATURES

The Automatic Irrigation System Based on Soil Moisture Content Control offers several advanced and useful features that make it efficient, reliable, and suitable for modern agricultural applications. One of the key features of this system is automatic irrigation control. The system operates without human intervention by continuously monitoring soil moisture levels and supplying water only when required. This eliminates the need for manual watering and reduces human effort.

Another important feature is soil moisture sensing. The system uses soil moisture sensors to accurately measure the water content present in the soil. Based on this real-time data, the system decides whether irrigation is needed or not, ensuring precise water management. The system also provides water conservation. By supplying water only when the soil moisture drops below a predefined threshold, it avoids over-irrigation and significantly reduces water wastage. This feature is especially beneficial in regions facing water scarcity.

A significant feature of the system is its microcontroller-based automation. The microcontroller (such as Arduino/ATmega328) acts as the brain of the system, processing sensor data and controlling the water pump accordingly. This ensures fast and reliable operation. The project supports energy efficiency as the water pump operates only when necessary, reducing unnecessary power consumption. This helps in lowering energy costs and improving overall system efficiency. Additionally, the system can be extended with IoT integration for remote monitoring and control. Farmers can monitor soil moisture levels and irrigation status through mobile applications or web platforms, making it more convenient and user-friendly. The system is also cost-effective and easy to implement, as it uses simple components and requires minimal maintenance. It can be used for different types of crops and soil conditions, making it a flexible solution. Finally, the system promotes sustainable agriculture by improving water management, reducing labor, and increasing crop yield. These features make it a smart and practical solution for modern farming needs.

2. LITERATURE SURVEY

2. LITERATURE SURVEY

1. **Design and Development of Automatic Irrigation System Based on Soil Moisture Sensor**
   Authors: Joaquín Gutiérrez, Juan F. Villa-Medina, Alejandra Nieto-Garibay, Miguel Ángel Porta-Gándara

   Description: This paper presents an automatic irrigation system that uses soil moisture sensors to control the water supply to crops. The system continuously monitors soil moisture levels and activates irrigation only when the moisture falls below a predefined threshold. By using microcontroller-based control, the system minimizes human intervention and reduces water wastage. Experimental results show improved water efficiency and healthier crop growth compared to traditional irrigation methods.

2. **Microcontroller-Based Automatic Plant Irrigation System Using Soil Moisture Sensors**
   Authors: Yunseop Kim, Robert G. Evans

   Description:
   The authors propose a low-cost, microcontroller-based irrigation system that automatically regulates irrigation based on real-time soil moisture data. Sensors are placed in the root zone to measure moisture content accurately. When the soil becomes dry, the controller activates the water pump and turns it off once optimal moisture is reached. The system improves irrigation accuracy and is suitable for small-scale and precision agriculture applications.

3. **Automatic Irrigation Control System Using Wireless Sensor Networks**
   Authors: L. Ruiz-Garcia, L. Lunadei, P. Barreiro, J. I. Robla

   Description: This research focuses on an automatic irrigation system that integrates soil moisture sensors with wireless sensor networks (WSNs). Sensor nodes collect moisture data from different agricultural zones and transmit it to a central controller. Irrigation decisions are made automatically based on moisture thresholds. The system enhances scalability, reduces wiring complexity, and supports efficient water management in large agricultural fields.

4. **Smart Irrigation System Based on Soil Moisture and Environmental Parameters**
   Authors: S. Harishankar, R. Sathish Kumar, K. P. Sudharsan, U. Vignesh

   Description: This paper introduces a smart irrigation system that uses soil moisture sensors along with environmental parameters such as temperature and humidity. The controller analyzes sensor data to determine irrigation needs and activates the irrigation system accordingly. The proposed method significantly reduces water consumption and ensures optimal soil conditions for crops, especially in water-scarce regions.

5. **Real-Time Automatic Irrigation System Using Soil Moisture Sensors and Embedded Technology**
   Authors: P. Divya, U. Umamakeswari

   Description: This study presents a real-time automatic irrigation system developed using embedded technology and soil moisture sensors. The system continuously monitors soil moisture and controls irrigation in real time without human involvement. It improves crop yield by maintaining proper soil moisture levels and reduces manual labor. The authors highlight its effectiveness in modern agriculture and its potential integration with IoT platforms.

2.1 REVIEW OF RELATED WORK

In recent years, significant research has been carried out in the field of automatic irrigation systems to improve water efficiency and agricultural productivity. Many systems have been developed using sensors, microcontrollers, and communication technologies to automate irrigation based on real-time conditions.

Earlier irrigation systems mainly relied on timer-based techniques, where water was supplied at fixed intervals regardless of the actual moisture condition of the soil. Although simple to implement, these systems were not efficient, as they often led to over-irrigation or under-irrigation. This resulted in water wastage, poor crop growth, and increased operational costs.

To overcome these limitations, researchers introduced soil moisture sensor-based irrigation systems. These systems measure the moisture level in the soil and activate irrigation only when the soil becomes dry. This approach significantly improved water management and ensured

that crops receive the required amount of water. Many studies have shown that such systems can reduce water consumption and improve crop yield.

Further advancements led to the development of microcontroller-based irrigation systems. In these systems, microcontrollers such as Arduino or ATmega328 are used to process sensor data and control the irrigation process automatically. These systems are reliable, easy to implement, and cost-effective, making them suitable for small and medium-scale farming.

With the growth of modern technology, wireless sensor networks (WSN) have been introduced in irrigation systems. In this approach, multiple sensors are deployed across the field to monitor different parameters such as soil moisture, temperature, and humidity. The collected data is transmitted wirelessly to a central system, which controls irrigation. This method improves coverage and accuracy but may increase system complexity and cost.

In recent years, IoT-based smart irrigation systems have gained popularity. These systems allow farmers to monitor and control irrigation remotely using mobile applications or web platforms. Real-time data monitoring, alerts, and data analysis help farmers make better decisions and improve overall farm management. However, IoT-based systems may require internet connectivity and higher initial investment.

Some research also focuses on combining multiple environmental parameters such as soil moisture, temperature, and weather conditions to make intelligent irrigation decisions. These systems provide better accuracy and efficiency but may require more sensors and complex programming. Despite these advancements, certain challenges still exist, such as high installation cost, maintenance issues, dependency on network connectivity, and lack of awareness among farmers. Therefore, there is a need for a simple, reliable, and cost-effective solution that can be easily adopted.

The proposed system addresses these issues by using a soil moisture sensor and microcontroller-based approach to automate irrigation. It provides an efficient solution by reducing water wastage, minimizing human effort, and improving crop productivity. The system is designed to be simple, affordable, and suitable for a wide range of agricultural applications.

2.2 DEFINITION OF PROBLEM STATEMENT

Traditional irrigation systems mainly depend on manual operation or fixed time schedules, which do not consider the actual moisture condition of the soil. This often leads to over-irrigation or under-irrigation, resulting in water wastage, increased energy consumption, and poor crop growth. In addition, variations in soil type, weather conditions, and crop requirements are not taken into account, making irrigation inefficient and labor-intensive. Therefore, there is a need for an automated irrigation system that can continuously monitor soil moisture levels and supply water only when required, ensuring efficient water usage, reduced human effort, and improved agricultural productivity.

2.3 EXISTING SYSTEM

The existing irrigation systems used in agriculture are mostly based on traditional methods such as manual irrigation and timer-based irrigation systems. In manual irrigation, farmers water the crops based on their experience and observation. This method requires continuous human effort and is not always accurate, as it depends on guesswork rather than actual soil conditions. As a result, crops may receive either excess water or insufficient water, affecting their growth and productivity.

Another commonly used method is the timer-based irrigation system, where water is supplied at fixed intervals of time. Although this method reduces manual effort, it does not consider real-time factors such as soil moisture, weather conditions, or crop requirements. This often leads to over-irrigation during rainy conditions or under-irrigation during dry periods. Consequently, a large amount of water is wasted, and energy consumption increases due to unnecessary operation of water pumps.

In many cases, traditional irrigation systems lack proper monitoring and control mechanisms. Farmers need to be physically present in the field to operate the system, which makes it labor-intensive and time-consuming. These systems also do not provide any feedback about soil conditions, making it difficult to maintain optimal moisture levels for crops. Moreover, conventional irrigation practices do not support modern technologies like automation, remote monitoring, or data analysis. This limits their efficiency and makes them unsuitable for precision agriculture. Due to these limitations, there is a need for an advanced irrigation system that can overcome these drawbacks by using real-time data and automated control.

Limitations of Existing System

The existing irrigation systems suffer from several limitations that reduce their efficiency and effectiveness in modern agriculture. One major drawback is the lack of real-time monitoring, as traditional methods do not consider the actual soil moisture condition before supplying water. This often leads to over-irrigation or under-irrigation, which negatively affects crop growth and results in water wastage.

Another limitation is the high dependency on manual labor. Farmers need to regularly monitor the fields and operate the irrigation system, which is time-consuming and labor-intensive. In timer-based systems, irrigation is done at fixed intervals without considering environmental factors such as rainfall, temperature, and soil type, making them inefficient.

Additionally, these systems cause unnecessary energy consumption because water pumps may run even when irrigation is not required. There is also no remote monitoring or control facility, which makes it difficult for farmers to manage irrigation from distant locations.

Overall, the existing systems lack automation, accuracy, and adaptability, making them unsuitable for efficient water management and modern precision agriculture.

2.4 PROPOSED SYSTEM

The proposed system is an automatic irrigation system that uses soil moisture sensors and a microcontroller to efficiently manage water supply to crops. The soil moisture sensor continuously monitors the moisture level in the soil and sends the data to the microcontroller. When the moisture level falls below a set threshold, the microcontroller automatically turns on the water pump to irrigate the field, and once the required moisture level is reached, it turns off the pump. This system ensures that water is supplied only when needed, reducing wastage and improving efficiency. It minimizes human intervention, saves energy, and supports better crop growth. The system is simple, cost-effective, and can also be enhanced with IoT features for remote monitoring and control, making it suitable for modern smart agriculture.

Advantages of the Proposed System:

The proposed automatic irrigation system offers several advantages over traditional irrigation methods. One of the main benefits is efficient water management, as the system supplies water only when the soil moisture level falls below the required threshold, thereby reducing water wastage. It also ensures proper irrigation**,** which helps in maintaining optimal soil conditions and improves crop growth and yield. Another important advantage is the reduction in manual effort. Since the system operates automatically, farmers do not need to constantly monitor the field or manually control irrigation, saving time and labor. The system also provides energy efficiency, as the water pump operates only when necessary, reducing electricity consumption.

The proposed system is cost-effective and easy to implement, making it suitable for both small-scale and large-scale farming. It is also reliable and accurate, as it uses real-time soil moisture data for decision-making. Additionally, the system can be enhanced with IoT technology for remote monitoring and control, providing greater convenience and flexibility. Overall, the system supports sustainable agriculture by conserving water, reducing resource usage, and improving productivity.

2.5 OBJECTIVES

The main objective of this project is to design and develop an automatic irrigation system that can efficiently manage water usage in agriculture by monitoring soil moisture levels. The system aims to provide water to crops only when required, ensuring optimal soil conditions for healthy plant growth.

Another objective is to reduce water wastage by avoiding over-irrigation and under-irrigation through real-time sensing and control. The project also focuses on minimizing human intervention by automating the irrigation process using a microcontroller and sensors.

Additionally, the system aims to improve agricultural productivity by maintaining proper moisture levels in the soil, which helps in better crop yield. It also targets energy efficiency by operating the water pump only when necessary. The project further aims to provide a cost-effective and easy-to-use solution that can be implemented in different agricultural environments. It also has the objective of supporting modern smart farming techniques, with the possibility of integrating IoT for remote monitoring and control in future enhancements.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

Hardware interfaces specify the physical and logical components required for the proper functioning of the system. The following are the hardware requirements used in this project:

- Processor : Arduino Uno (ATmega328 Microcontroller)
- Soil Moisture Sensor
- Water Pump / Motor
- Relay Module
- Power Supply (5V/9V Battery or Adapter)
- Connecting Wires
- LCD Display (Optional)
- Breadboard

2.6.2 SOFTWARE REQUIREMENTS:

Software requirements specify the programs and tools used to design, develop, and control the system. The following are the software requirements:

- Operating System : Windows 10
- Programming Language : Embedded C / Arduino C
- Software Tool : Arduino IDE
- Simulation Tool (Optional) : Proteus
- Libraries Used : Arduino Standard Libraries

3. SYSTEM ARCHITECTURE & DESIGN

3.SYSTEM ARCHITECTURE & DESIGN

Project architecture refers to the structural framework and design of a project, including its components, interactions, and overall organization. It provides a clear blueprint for the development of the system, ensuring efficiency, reliability, and proper functioning. A well-designed architecture helps in simplifying the implementation process, improving system performance, and supporting future enhancements.

3.1 PROJECT ARCHITECTURE

This project architecture represents the working flow of the Automatic Irrigation System, from sensing soil moisture to controlling the water pump.
It follows the process of input (sensor) → processing (microcontroller) → output (irrigation system).

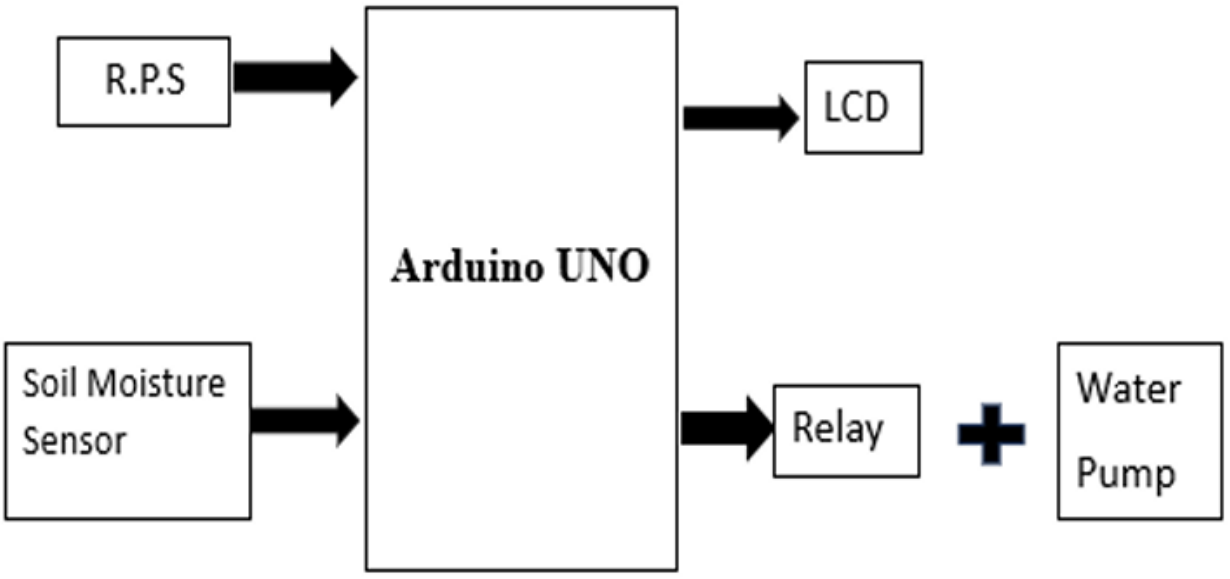

Figure 3.1: Project Architecture

3.2 DESCRIPTION

Input Data : The project uses real-time soil data, specifically soil moisture levels obtained from the soil moisture sensor placed in the agricultural field.

Reading Data : The soil moisture sensor continuously reads the moisture content in the soil and sends the data to the microcontroller for processing.

Feature Extraction : The sensor converts soil moisture levels into electrical signals (analog/digital values), which are used as input features for decision-making.

Processing : The microcontroller (Arduino/ATmega328) processes the sensor data by comparing it with a predefined threshold value to determine the soil condition.

Decision Mechanism : Based on the processed data, the system decides whether irrigation is required or not. If the soil is dry, it triggers the irrigation system.

Control Mechanism : A relay module is used to control the water pump. The microcontroller sends signals to the relay to turn the pump ON or OFF.

Output : The system automatically supplies water to the crops when needed and stops irrigation once the required moisture level is reached.

Monitoring : The system continuously monitors soil conditions in real-time, ensuring efficient and timely irrigation.

Feedback : The system uses continuous sensor feedback to adjust irrigation automatically, improving accuracy and maintaining optimal soil moisture levels.

3.3 DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a graphical representation that illustrates how data flows within the Automatic Irrigation System, showing its processes, data inputs, outputs, and storage. It is an important tool in system design that helps in understanding how the system operates and how information moves between components.

A Data Flow Diagram comprises four primary elements:

• **External Entities:** These represent sources or destinations of data outside the system, such as the soil environment and user (farmer).

• **Processes:** These indicate operations performed on data, such as reading soil moisture, processing data in the microcontroller, and controlling the water pump.

• **Data Flows:** These show the movement of data between components, such as moisture data from the sensor to the microcontroller and control signals to the relay.

• **Data Stores:** These represent where data is stored within the system, such as threshold values or system parameters in the microcontroller memory.

These components are represented using standard symbols, such as circles for processes, arrows for data flow, rectangles for external entities, and open-ended rectangles for data storage.

Benefits:

The visual representation of the DFD makes it easy to understand for both technical and non-technical users. It helps in identifying how data is processed within the irrigation system and ensures efficient system design. It also improves communication among developers and helps in detecting errors or inefficiencies in the workflow.

Applications:

DFDs are used in designing embedded systems, agricultural automation systems, and IoT-based applications. In this project, the DFD helps in understanding how soil moisture data is collected, processed, and used to control irrigation automatically.

Levels of DFD:

DFDs are structured hierarchically:

• **Level 0 (Context Diagram):** Provides a high-level overview of the Automatic Irrigation System, showing interaction between the system, soil, and user.

• **Level 1:** Breaks down the main process into sub-processes such as moisture sensing, data processing, and pump control.

• **Level 2+:** Provides detailed representation of individual processes like sensor operation and decision-making logic.

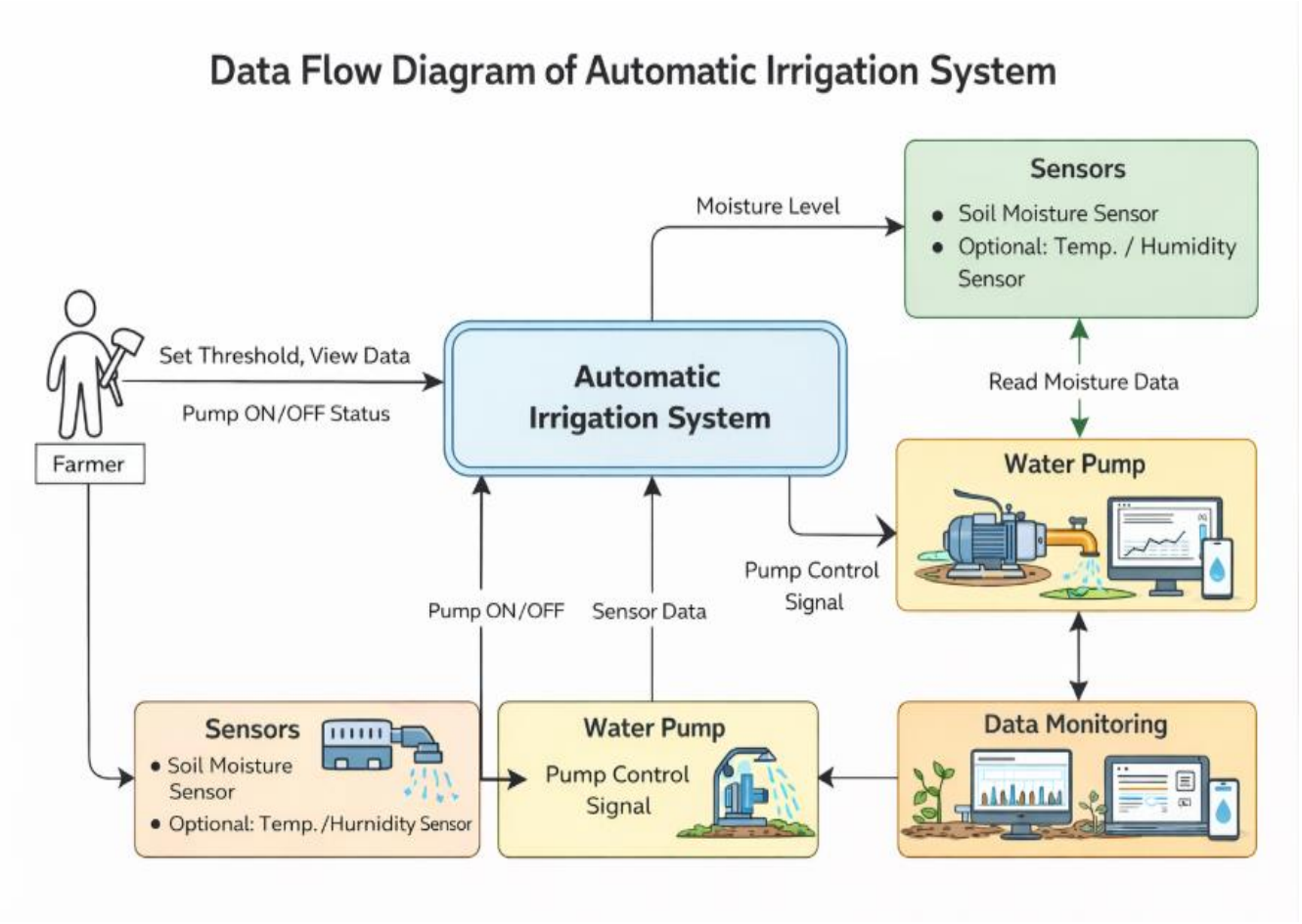

Figure 3.2: Data flow Diagram

4. IMPLEMENTATION

4. IMPLEMENTATION

The implementation phase of the Automatic Irrigation System involves integrating hardware and software components to achieve efficient and automated irrigation. This phase focuses on assembling the required components, programming the microcontroller, and ensuring proper coordination between all modules.

The system is implemented using a soil moisture sensor, Arduino Uno (ATmega328 microcontroller), relay module, and water pump. The soil moisture sensor is placed in the soil to measure the water content continuously. The sensor outputs analog signals based on the moisture level, which are sent to the Arduino for processing.

The Arduino microcontroller acts as the core of the system. It is programmed using Arduino IDE to read sensor values, process the data, and control the irrigation system. The program continuously checks the soil moisture level and compares it with a predefined threshold value.

When the moisture level falls below the threshold, the Arduino sends a signal to the relay module. The relay acts as a switch and turns ON the water pump, allowing water to flow to the crops. Once the soil moisture reaches the required level, the Arduino automatically turns OFF the pump. This process runs continuously in a loop, ensuring real-time monitoring and control. The implementation is simple, cost-effective, and reliable. It reduces manual effort and ensures efficient water usage. Additional features such as LCD display or IoT modules can be integrated for real-time monitoring and remote control.

4.1 ALGORITHMS USED

The Automatic Irrigation System uses simple decision-making and control algorithms to automate the irrigation process based on soil moisture conditions.

1. Threshold-Based Control Algorithm

The primary algorithm used in this system is a threshold-based decision algorithm.

Working:

- The system continuously reads soil moisture values from the sensor.
- These values are compared with a predefined threshold level.
- If the moisture level is below the threshold (dry soil), irrigation is required.
- If the moisture level is above the threshold (wet soil), irrigation is stopped.

Advantages:

- Simple and easy to implement
- Requires less computational power
- Provides quick response

Disadvantages:

- Fixed threshold may not suit all soil types
- Does not consider environmental factors like temperature or humidity

2. Microcontroller-Based Control Logic

The microcontroller follows a continuous loop-based control algorithm.

Working:

- Initialize sensor and relay module
- Read moisture value from sensor
- Process the data inside the microcontroller
- Send control signals to relay
- Repeat the process continuously

Advantages:

- Real-time monitoring and control
- Reliable and efficient
- Reduces human intervention

Disadvantages:

- Limited processing capability compared to advanced systems
- Requires proper calibration

3. Relay Switching Algorithm

The relay module works based on digital signals from the microcontroller.

Working:

- If signal = HIGH → Relay ON → Pump ON
- If signal = LOW → Relay OFF → Pump OFF

Advantages:

- Simple ON/OFF control
- Low power consumption
- Easy integration with microcontroller

Disadvantages:

- Only supports binary control (no speed control of pump)

4. Feedback Mechanism

The system uses a continuous feedback loop for better performance.

Working:

- After irrigation, soil moisture level changes
- Sensor detects updated moisture level
- Data is sent again to microcontroller
- System adjusts irrigation accordingly

Advantages:

- Maintains optimal soil moisture
- Improves system accuracy
- Prevents over-irrigation

OVERALL SYSTEM WORKING

Now-a-days, efficient water management is essential due to increasing water scarcity and agricultural demands. Traditional irrigation methods waste water and require manual effort. To overcome this, the proposed system uses a soil moisture sensor and microcontroller-based automation.

The system continuously monitors soil moisture and automatically controls irrigation. When the soil becomes dry, the system activates the water pump, and when sufficient moisture is reached, it stops irrigation. This ensures proper water usage and improves crop growth.

The proposed system is simple, cost-effective, and suitable for real-time applications. It can be further enhanced with IoT for smart agriculture. Compared to traditional methods, this system provides better efficiency, reduces labor, and supports sustainable farming practices.

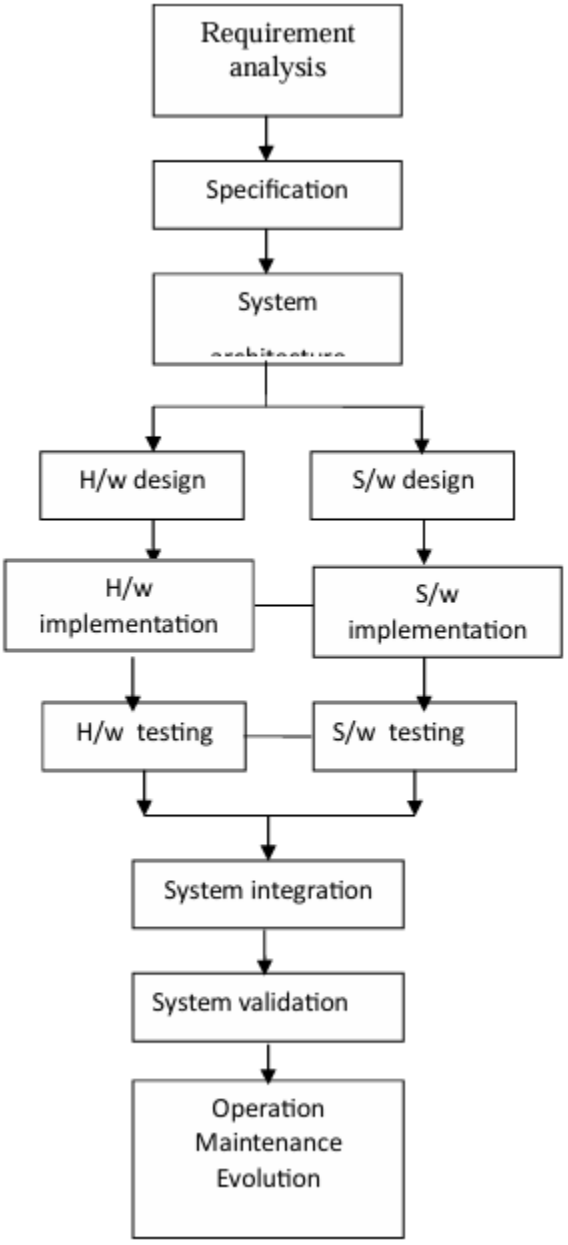

Figure 4.1: Embedded Development Life Cycle

4.2 SAMPLECODE

```
#include <LiquidCrystal.h>
#include <stdio.h>

LiquidCrystal lcd(6, 7, 5, 4, 3, 2);
// defines pins numbers
//New GPS GY-GPS6MV2

int mos    = 8;
int pump   = 9;
int buzzer = 13;



int tempc=0;

int hbtc=0,hbtc1=0,rtrl=0,rtr2=0;
int temps=0,hums=0,alcs=0,eyes=0,buttons=0;

 unsigned char rcv,count,gchr='x',gchr1='x',robos='s';

 char rcvmsg[10],pastnumber[11];

 char gpsval[50];
// char dataread[100] = "";
// char lt[15],ln[15];

int i=0,k=0,lop=0;
int  gps_status=0;
float latitude=0;
float logitude=0;
String Speed="";
String gpsString="";
char *test="$GPRMC";
```

```
//int hbtc=0,hbtc1=0,rtrl=0;

unsigned char gv=0,msg1[10],msg2[11];
 float lati=0,longi=0;
unsigned int lati1=0,longi1=0;
unsigned char flat[5],flong[5];
unsigned char finallat[8],finallong[9];

 int ii=0,rchkr=0;

String inputString = "";         // a string to hold incoming data
boolean stringComplete = false;  // whether the string is complete

void okcheck()
{
  unsigned char rcr;
  do{
      rcr = Serial.read();
    }while(rcr != 'K');
}

 void sound()
 {
  digitalWrite(buzzer,LOW);delay(1500);digitalWrite(buzzer,HIGH);
 }

void setup()
{
 // unsigned char rcr;
 Serial.begin(9600);//serialEvent();

  pinMode(mos, INPUT);pinMode(pump, OUTPUT);
  pinMode(buzzer, OUTPUT);
```

```
  digitalWrite(pump, LOW);digitalWrite(buzzer, HIGH);

  lcd.begin(16, 2);lcd.cursor();
  lcd.print("Aut Irrigation");
  lcd.setCursor(0,1);
  lcd.print("Soil Mos Ctrl");
      delay(2500);

    lcd.clear();
    lcd.setCursor(0,0);
    lcd.print("Mos:");    //4,0
}

void loop()
{
  if(digitalRead(mos) == LOW)
    {
      lcd.setCursor(4,0);lcd.print("Wet_Pump_OFF");
      digitalWrite(pump, LOW);
    }
  if(digitalRead(mos) == HIGH)
    {
      lcd.setCursor(4,0);lcd.print("Dry_Pump_ON ");
      digitalWrite(pump, HIGH);
    }
  delay(800);
}
```

5. RESULTS & DISCUSSION

5.RESULTS&DISCUSSION

The Automatic Irrigation System Based on Soil Moisture Content Control was successfully implemented and tested under different soil conditions. The system effectively monitored the soil moisture level and automatically controlled the irrigation process without any manual intervention.

During testing, the soil moisture sensor accurately detected the moisture content in the soil and sent real-time data to the microcontroller. When the soil was dry, the system automatically activated the water pump through the relay module. Once the soil reached the required moisture level, the system turned OFF the pump. This demonstrates that the system works efficiently and responds correctly to changing soil conditions.

The results show that the system helps in **reducing water wastage** by supplying water only when necessary. It also minimizes **human effort** and ensures proper irrigation, leading to better crop growth. The system was found to be reliable, cost-effective, and suitable for real-time applications.

In addition, the system provides quick response and continuous monitoring, which improves overall irrigation efficiency. The performance of the system proves that it is a practical solution for modern agriculture and can be further enhanced using IoT for remote monitoring.

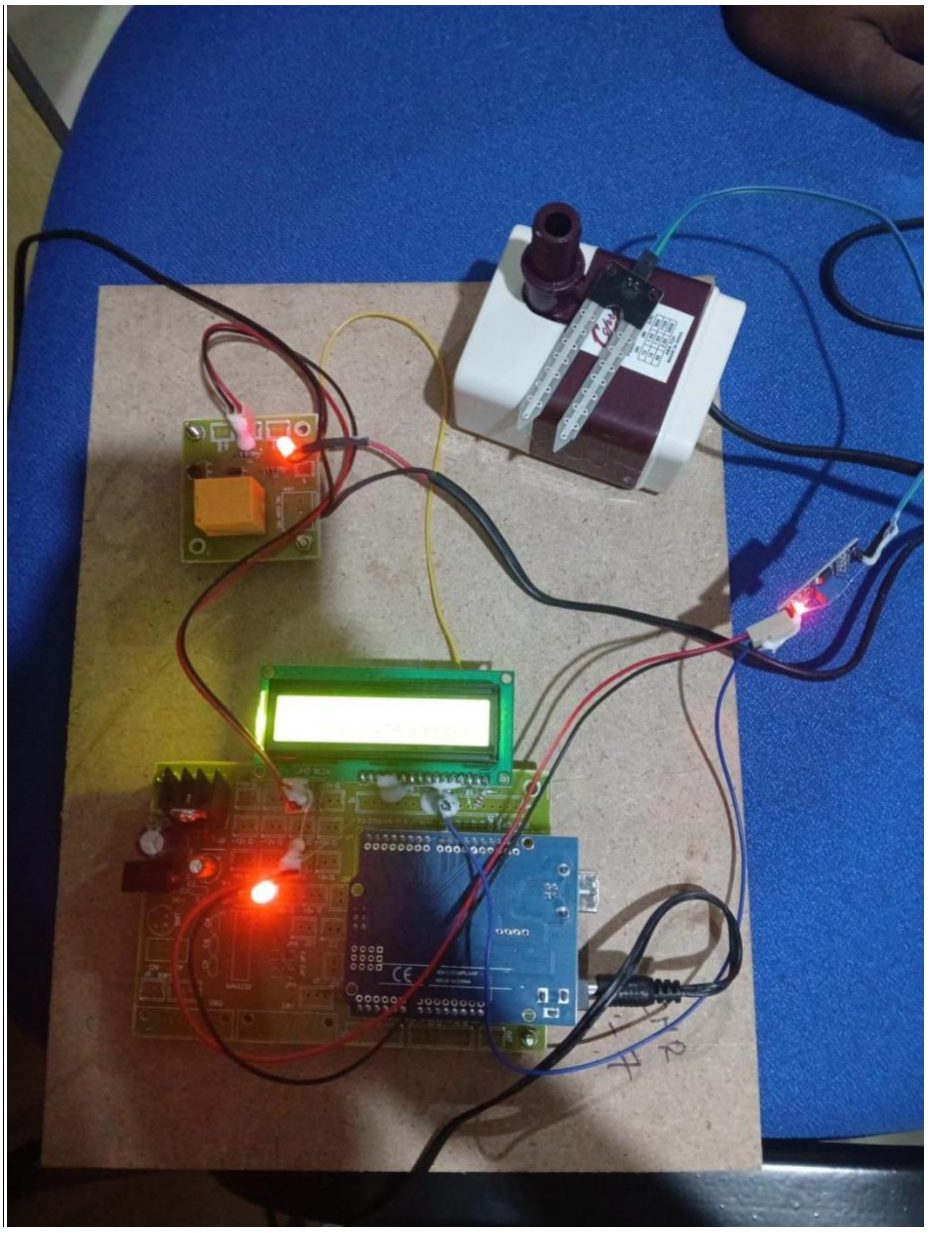

Figure 5.1: Hardware Setup of Automatic Irrigation System

6.VALIDATION

6.1 VALIDATION

Validation is an important phase of the project that ensures the system works correctly according to the requirements and produces expected results. In the Automatic Irrigation System Based on Soil Moisture Content Control, validation is performed by testing the system under different soil moisture conditions and verifying its response.

The system was tested by placing the soil moisture sensor in both dry and wet soil conditions. When the soil was dry, the sensor detected low moisture levels and sent signals to the microcontroller. The microcontroller correctly processed this data and activated the relay module, which turned ON the water pump. Similarly, when the soil was sufficiently wet, the sensor detected high moisture levels, and the microcontroller turned OFF the pump automatically. This confirms that the system responds accurately based on real-time soil conditions.

Multiple test cases were conducted to validate the reliability of the system. The system consistently performed as expected, with quick response time and accurate decision-making. The threshold value was adjusted and tested to ensure proper irrigation without overwatering or underwatering. This helped in fine-tuning the system for better performance.

The validation process also ensured that all hardware components, including the soil moisture sensor, Arduino microcontroller, relay module, and water pump, were functioning properly and were well integrated. The system showed stable operation without any major errors during continuous testing.

Additionally, the system was evaluated for efficiency in terms of water usage and energy consumption. It was observed that water was supplied only when required, reducing wastage significantly. The pump operated only when necessary, which helped in saving electrical energy.

Overall, the validation results confirm that the proposed system is reliable, efficient, and suitable for real-time agricultural applications. The system meets the project objectives by automating irrigation, conserving water, and reducing manual effort. Further improvements can be made by integrating advanced technologies like IoT for remote monitoring and control.

6.2 TESTCASES

TABLE 6.2.1 SOIL MOISTURE DETECTION

Testcase ID	Testcase Name	Purpose	Test Case	Output
1	Soil Moisture Detection	To check sensor reading	Sensor is placed in soil	Moisture value displayed/read successfully

TABLE 6.2.2 PUMP ACTIVATION (DRY SOIL)

Testcase ID	Testcase Name	Purpose	Input	Output
1	Pump ON Test	To check pump activation	Dry soil condition (low moisture)	Pump turns ON

TABLE 6.2.3 PUMP DEACTIVATION (WET SOIL)

Testcase ID	Testcase Name	Purpose	Input	Output
1	Pump OFF Test	To check pump deactivation	Wet soil condition (high moisture)	Pump turns OFF

TABLE 6.2.4 CONTINUOUS MONITORING

Testcase ID	Testcase Name	Purpose	Input	Output
1	Continuous Monitoring Test	To verify real-time operation	Sensor continuously in soil	System monitors and controls pump automatically

TABLE 6.2.5 SYSTEM RESPONSE

Testcase ID	Testcase Name	Purpose	Input	Output
1	Response Time Test	To check system response	Change in soil moisture	Immediate ON/OFF action of pump

7.CONCLUSION
&
FUTURE ASPECTS

7.CONCLUSION & FUTUREASPECTS

In conclusion, the Automatic Irrigation System Based on Soil Moisture Content Control has been successfully designed and implemented. The system effectively automates the irrigation process by monitoring soil moisture levels and supplying water only when required. The implementation was carried out efficiently, ensuring proper coordination between hardware and software components. This project helps in reducing water wastage, minimizing human effort, and improving agricultural productivity. The results demonstrate that the system is reliable, cost-effective, and suitable for real-time applications.

Looking ahead, the project has great potential for further enhancements. Future developments can focus on integrating advanced technologies such as IoT, wireless communication, and smart sensors. These improvements will enhance system efficiency, enable remote monitoring, and support sustainable agriculture practices. Such advancements will ensure that the system remains relevant and beneficial for modern farming needs.

7.1 PROJECT CONCLUSION

This project presents an effective and practical solution for automated irrigation using soil moisture sensing and microcontroller-based control. The system continuously monitors soil conditions and makes real-time decisions to control the water pump, ensuring optimal irrigation. By using a simple threshold-based algorithm and reliable hardware components, the system successfully reduces water wastage and energy consumption.

The proposed system overcomes the limitations of traditional irrigation methods by providing automation, accuracy, and efficiency. It ensures that crops receive the required amount of water at the right time, leading to improved crop growth and yield. The system is easy to implement, affordable, and adaptable to different agricultural environments.

Overall, this project contributes to smart agriculture by promoting efficient resource utilization and reducing dependency on manual labor. It demonstrates how embedded systems can be effectively used to solve real-world agricultural problems.

7.2 FUTUREASPECTS

The proposed Automatic Irrigation System has shown promising results in improving irrigation efficiency. However, there are several opportunities for further development to enhance its performance and applicability in real-world scenarios.

Some future aspects of the project include:

• **IoT Integration:** Enabling remote monitoring and control of the system through mobile applications or web platforms.
• **Weather-Based Irrigation:** Incorporating weather data (rainfall, temperature, humidity) for smarter irrigation decisions.
• **Solar Power Integration:** Using solar panels to make the system energy-efficient and eco-friendly.
• **Advanced Sensors:** Using multiple sensors (temperature, humidity, water level) for better accuracy.
• **Automatic Fertilizer System:** Integrating fertigation systems to supply nutrients along with water.
• **Wireless Communication:** Implementing GSM/Wi-Fi modules for real-time alerts and updates.
• **Scalability:** Expanding the system for large-scale farming and smart agriculture applications.
• **AI-Based Optimization:** Applying machine learning techniques to predict irrigation needs more accurately.

8.BIBLIOGRAPHY

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8.2 GITHUB LINK

https://github.com/Manojkumar128/Automatic-irrigation-soil-moisture-content-control.git